

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons) — Semester V Examination, 2022-23

Paper No. MTB-509: Special Theory of Relativity-I (2012)

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1.

- (a) Show that Newton's second law of motion is invariant under special Galilean transformations. [4]
- (b) Show that the Lorentz transformation equations are superior to Galilean transformations. [4]
- (c) At what speed should a clock be moved so that it may appear to lose 1 minute in each hour? [2]
- (d) The length of a rocket ship on the ground is 100 m. When it is in flight, its length observed from the ground is 99 m. Calculate its speed. [2]
- (e) Show that if V_0 is the rest volume of a cube, then $V = V_0 \sqrt{1 - v^2/c^2}$ is the volume viewed from a frame moving parallel to an edge of the cube. [2]
- (f) Show that the circle $x^2 + y^2 = a^2$ in frame S appears as an ellipse in frame S' moving with velocity v along the x -axis. [3]
- (g) Explain Minkowski four-dimensional space-time. [3]
- (h) Define the 4-dimensional flat space-time metric. What is its use for 4-D vectors? [2]

Question 2.

- (a) A Cartesian frame $OX'YZ$ rotates about the axis OZ with uniform angular velocity ω relative to the inertial frame $OXYZ$. Show that $OX'YZ$ is non-inertial. [6]
- (b) Prove that the Lorentz transformation equations can be written in the matrix form:

$$\begin{pmatrix} x' \\ ct' \end{pmatrix} = \begin{pmatrix} \cosh \theta & -\sinh \theta \\ -\sinh \theta & \cosh \theta \end{pmatrix} \begin{pmatrix} x \\ ct \end{pmatrix}$$

Hence show that Lorentz transformations can be regarded as a rotation through an imaginary angle. [6]

Question 3.

- (a) Prove that the set of all Lorentz transformations satisfies the group properties. [6]
- (b) Determine the component velocities of a particle with respect to two different frames S and S' using the Lorentz transformation (relativistic velocity addition). [6]

Question 4.

- (a) By using Lorentz transformation equations, show that simultaneity is not absolute but relative. [6]
- (b) Calculate the length of a rod moving with velocity $0.8c$ in a direction inclined at 60° to its own length. The proper length is 100 cm. [6]

Question 5.

- (a) Prove that the three-dimensional volume element $dx dy dz$ is not Lorentz invariant, but the four-dimensional volume element $dx dy dz dt$ is invariant under Lorentz transformation. [6]
- (b) If u and u' are velocities of a particle in two inertial systems S and S' (where S' has velocity v relative to S in the x -direction), show that:

$$u^2 = \frac{u'^2 + v^2 + 2u'v \cos \theta' - \left(\frac{u'v \sin \theta'}{c}\right)^2}{\left(1 + \frac{u'v \cos \theta'}{c^2}\right)^2}$$

where θ' is the angle u' makes with the x -axis. [6]

Question 6.

- (a) Find the velocity at which the mass of a particle is double its rest mass. [5]
- (b) Explain the Michelson-Morley experiment. [7]

Question 7. Explain the following terms in detail: (a) Light cone; (b) World points and world lines; (c) Space-like intervals; (d) Time-like intervals. [12]

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